Building Damage Assessment using Drone Imagery

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Background and motivation

The challenge of post-disaster response

Natural disasters — earthquakes, floods, and hurricanes — can cause severe and widespread damage to buildings and infrastructure. Rapidly assessing that damage is critical for guiding rescue teams and prioritising recovery efforts.

AI and satellite imagery have limitations

AI-based approaches using satellite datasets like xBD have shown promise, but satellite imagery is captured from very high altitudes — making it difficult to detailed structural damage at the building level.

Why drone imagery?

Government agencies increasingly deploy drones (UAVs) in real disaster response. Drones can be mobilised quickly and capture high-resolution, close-range imagery, offering far richer visual detail than satellite alternatives.

Problem Statement

- After a disaster, it is important to quickly identify **which buildings are damaged.**
- Currently, this is often done by **manually analyzing drone images**, where experts look at the images and check buildings one by one.
- This process **takes a lot of time and effort,** especially when there are **many buildings in large affected areas.**
- Therefore, there is a need for an **automated system that can analyze aerial images and detect damaged buildings more quickly**, helping disaster response teams understand the situation faster.

Existing Work

Recent research has explored the use of **AI with drone imagery for building damage assessment**.

1. The **CRASAR-U-DROIDs dataset paper (2024)** introduces a large **UAV dataset with labeled buildings and damage information**, which supports training AI models for detecting damage from drone images.
2. Another study focuses on the **deployment of AI systems during real disaster response,** showing that AI can analyze large amounts of drone imagery collected after disasters. The main objective of this work was to **demonstrate that AI-based damage assessment using UAV imagery can be applied in real-world disaster situations,** rather than proposing a new model or detailed training process.
3. These **two papers form the base of our research,** highlighting the importance of **drone imagery and the potential of AI for faster damage assessment.**

1. Dataset Paper - Introduces a large UAV dataset with labeled building damage.
2. Deployment Paper - Recent research has explored the use of AI with drone imagery for building damage assessment.

Research Gap

1. Most existing building damage research relies on satellite imagery (e.g., xBD), which lacks detailed structural information due to lower resolution.
2. The CRASAR-U-DROIDs UAV dataset is new and has only been tested with basic baseline models, leaving room for more advanced AI methods.

Project Objective

Develop an **AI-based system to classify and localize building damage from drone imagery**, enabling faster assessment of disaster-affected areas.

Input Data

We use the **CRASAR-U-DROIDs dataset**, which contains drone images **collected from real disaster events.**

GeoTiff files

1. **GeoTIFF image**: an image where **each pixel is linked to a real-world geographic location.**
2. Collected from **10 real disaster events** including hurricanes, wildfire, tornado, volcanic eruption, and building collapse
3. Contains **52 orthomosaic drone images**
4. Includes **21,716 labeled buildings with damage annotations**

Annotation Files

1. **JSON file:** contains the **location of each building in the image and its damage label**
2. **Aligned JSON file:** contains **corrected building annotations so that the building outlines match the imagery properly,** since buildings may get **shifted or transformed when multiple drone images are stitched together.**

Dataset Link : https://huggingface.co/datasets/CRASAR/CRASAR-U-DROIDs

Initial Approach

Initially, we tried direct building damage classification.

1. Extracted building patches using the JSON annotations
2. Each building crop was given as input to the model
3. Models used:
   CNN
   ResNet
4. The task was to classify each building into damage categories.

Initial Model Results

1. CNN (Musset dataset): Macro-F1 = 0.37
2. ResNet (Musset dataset): Macro-F1 = 0.65
3. ResNet (Laura + Eruption): Macro-F1 = 0.58

```
•••           precision    recall  f1-score   support

           0       0.67      0.22      0.33         9
           1       0.00      0.00      0.00         1
           2       0.47      0.88      0.61         8

    accuracy                           0.50        18
   macro avg       0.38      0.37      0.31        18
weighted avg       0.54      0.50      0.44        18

[[2 0 7]
 [0 0 1]
 [1 0 7]]
```

```
FINAL TEST RESULTS
========================================
Accuracy: 0.9118

Classification Report:
              precision    recall  f1-score   support

           0     0.9667    0.9355    0.9508        31
           1     0.0000    0.0000    0.0000         1
           3     0.6667    1.0000    0.8000         2

    accuracy                         0.9118        34
   macro avg     0.5444    0.6452    0.5836        34
weighted avg     0.9206    0.9118    0.9140        34

Confusion Matrix:
[[29  1  1]
 [ 1  0  0]
 [ 0  0  2]]
```

```
FINAL TEST RESULTS
========================================
Accuracy: 0.8640

Classification Report:
              precision    recall  f1-score   support

           0     0.9049    0.9407    0.9224       455
           1     0.3636    0.2791    0.3158        43
           2     1.0000    0.1111    0.2000        18
           3     0.7778    1.0000    0.8750        28

    accuracy                         0.8640       544
   macro avg     0.7616    0.5827    0.5783       544
weighted avg     0.8587    0.8640    0.8481       544

Confusion Matrix:
[[428  20   0   7]
 [ 30  12   0   1]
 [ 15   1   2   0]
 [  0   0   0  28]]
```

Limitations of the Initial Approach

1. In real scenarios, **only drone images are available**, not JSON files
2. Drone images are **stitched together to form orthomosaics**
3. This can cause **misalignment between the annotations and the actual buildings**

So relying only on **JSON annotations is not reliable for real-world use**.

Updated Approach

To overcome this, we changed the approach. We use a **two-stage pipeline:**

1. **Localization** – detect buildings directly from the image
2. **Classification** – classify the detected buildings into damage categories

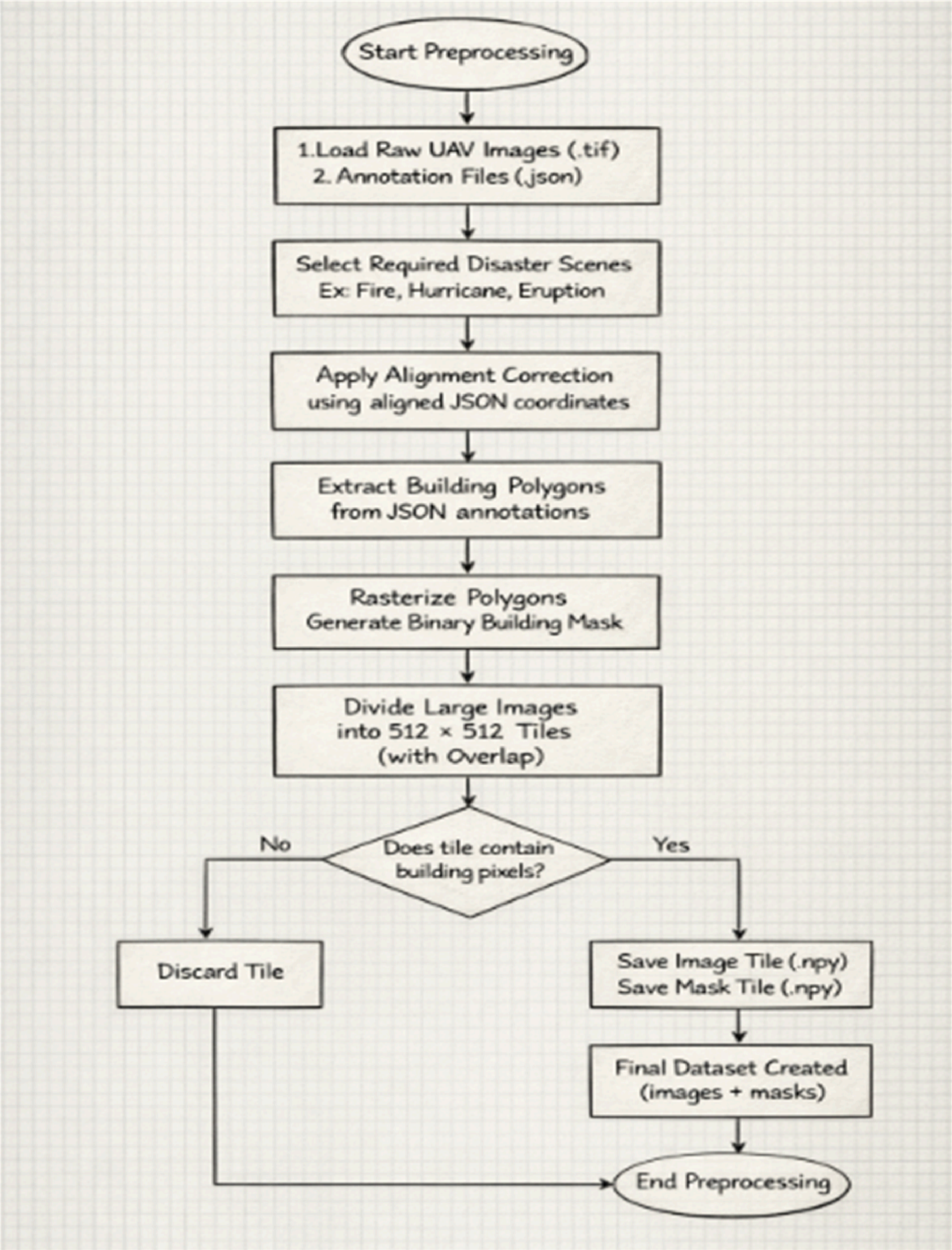
This method works directly with drone images, making it more practical for real-world disaster analysis.

Building Localization

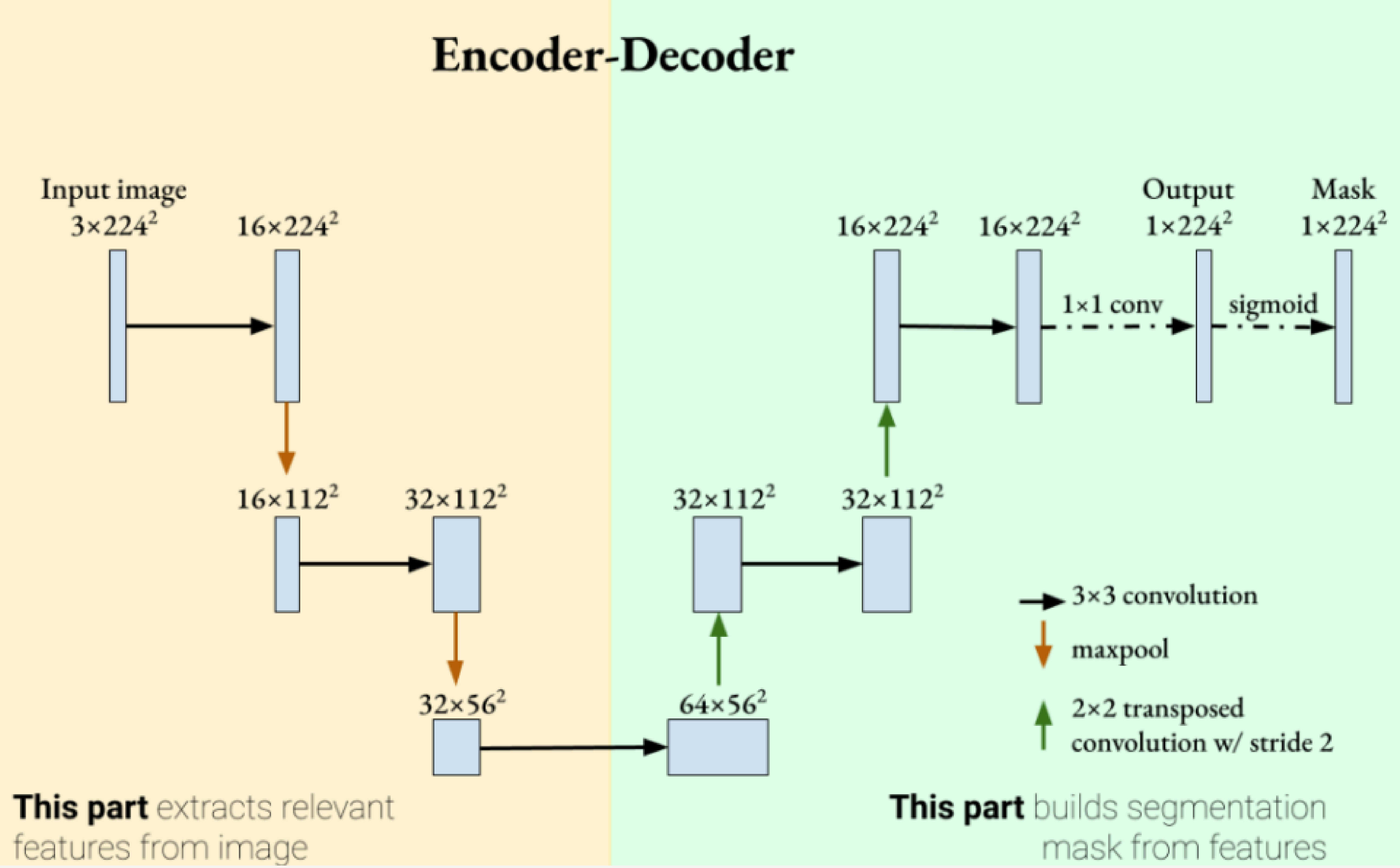
Building localization is the **first step of our system**.
It focuses on **identifying where buildings are present in the drone image**.

1. Detects **building regions directly from the drone images**
2. Helps **locate individual buildings in large aerial images**
3. The detected building areas are then **extracted and used for damage classification**

Data Preprocessing



U-Net Architecture for Building Localization



Model 0: Initial Segmentation Experiment

In **Model 0**, we trained a basic **U-Net segmentation model** to learn the task of **building localization** from drone imagery.

1. **Loss Function:** Dice Loss
2. **Goal:** Establish a baseline model for detecting building regions.

However, the dataset is **highly imbalanced**, where most pixels belong to the **background** and only a small portion represent **buildings**. Because Dice Loss alone could not properly handle this imbalance, the model often **predicted most pixels as background**.

This experiment showed that **Dice Loss alone is not sufficient for this dataset**, indicating the need to explore **loss functions that better handle class imbalance**.

Model	**Dataset**	**Activation**	**Loss**	**Dice**	**IoU**	**Accuracy**
U-Net (CNN Encoder-Decoder Semantic Segmentation)	Mussett Bayou Fire	Sigmoid	Dice Loss	0.47	0.31	0.31

Model 1: Improved Segmentation Model

After observing the limitations of Model 0, we improved the model by **changing the encoder architecture and modifying the loss function.**

1. **Encoder Change:** Basic CNN encoder → **ResNet-34 encoder**
2. **Loss Function:** Combination of **Binary Cross Entropy (BCE) and Dice Loss**
3. **Dataset Split**: 80% training and 20% testing

Using **ResNet-34 as the encoder** helped the model learn stronger visual features such as **edges, textures, and building structures,** because it is a deeper and pretrained architecture.

Combining **BCE with Dice Loss** also helped the model better distinguish **building pixels from background pixels,** addressing the imbalance observed in Model 0.

This experiment produced **more stable building predictions** and established a stronger baseline for further experiments.

Model	Dataset	Activation	Loss	Dice	IoU	Accuracy
U-Net (ResNet-34 Encoder)	Mussett Bayou Fire	Sigmoid	BCE + Dice Loss	0.62	0.45	0.74

Model 2.1: Geographic Split Experiment

In the previous model, the dataset was split **randomly into training and testing sets.** However, random splitting can place **similar image tiles from the same area in both training and testing,** which can lead to overly optimistic results.
To address this, we introduced a **geographic split strategy.**

1. **Training Areas:** MussettBayouFire-03, 04, 05
2. **Testing Areas:** MussettBayouFire-01, 02
3. **Loss Function: Tversky Loss**

This ensures that the model is tested on **different geographic regions that were not seen during training,** making the evaluation more realistic.
Since the dataset still contained many **background-only tiles,** we also applied **balanced sampling during training:**

1. **70% tiles containing buildings**
2. **30% background tiles**

Using **Tversky Loss** helped the model better handle the **class imbalance between building pixels and background pixels,** improving building detection performance.

Model	Dataset	Activation	Loss	Dice	IoU	Accuracy
U-Net (ResNet-34 Encoder + Balanced + Spatial Split)	Mussett Bayou Fire	Sigmoid	Tversky Loss	0.67	0.51	0.79

Model 2.2: Multi-Disaster Training

In Model 2.1, the model was trained and tested on **different geographic regions of the same disaster. However, buildings from the same disaster** often share **similar visual patterns,** which may still limit the model's ability to generalize.

To address this, we expanded the training data to include **multiple disasters.**

1. **Training Data:** Mussett Bayou Fire, Hurricane Laura, and Volcanic Eruption scenes
2. **Model:** U-Net with **ResNet-50 encoder**
3. **Goal:** Learn building features from **different disaster environments**

Training on multiple disasters allows the model to observe **more diverse building structures, damage patterns, and environmental conditions,** helping it learn **more general building features** instead of patterns specific to a single disaster.

Model	**Dataset**	**Activation**	**Loss**	**Dice**	**IoU**	**Accuracy**
U-Net (ResNet-50 Encoder)	Mussett Fire, Hurricane Laura, Volcanic Eruption	Sigmoid	Tversky Loss	0.62	0.45	0.78

Model 3.1: Cross-Disaster Generalization (ResNet Encoder)

To evaluate whether the model can detect buildings in **completely unseen disaster events,** we designed a **cross-disaster experiment.**

1. **Training Disasters**: Hurricane Laura and Hurricane Ian
2. **Testing Disasters**: Hurricane Michael and Hurricane Idalia
3. **Model:** U-Net with **ResNet encoder**

This setup ensures that the model is evaluated on **disasters that were not part of the training data,** providing a more realistic assessment of real-world performance.

To improve the model's robustness, we also applied **data augmentation techniques** such as flipping, rotation, and brightness adjustments. These augmentations help the model learn **building patterns under different environmental and imaging conditions.**

Model	**Dataset**	**Activation**	**Loss Function**	**Dice Score**	**IoU**	**Overall Accuracy**
U-Net (ResNet-50 Encoder + SCSE Attention)	Hurricane UAV scenes (Laura & Ian → Train, Michael & Idalia → Test)	Sigmoid	Focal Tversky Loss	0.67	0.5	0.75

Model 3.2: EfficientNet Encoder Experiment

After evaluating cross-disaster performance using a **ResNet encoder**, we explored whether **a stronger feature extractor** could further improve building detection.

1. **Encoder Change:** ResNet → **EfficientNet-B4**
2. **Loss Function: Focal Tversky Loss**
3. **Training Setup:** Same cross-disaster split as Model 3.1

EfficientNet provides **more powerful feature extraction**, allowing the model to learn **complex building structures across different disasters.**

Focal Tversky Loss further improves training by **focusing on difficult building pixels**, which helps handle the **strong class imbalance present in the dataset.**

This experiment evaluates whether **a stronger encoder combined with improved loss weighting can enhance generalization to unseen disasters.**

Model	**Dataset**	**Activation**	**Loss Function**	**Dice Score**	**IoU**	**Overall Accuracy**
U-Net (EfficientNet-B4 Encoder + SCSE Attention)	Hurricane UAV scenes (Ian & Laura → Train, Idalia & Michael → Test)	Sigmoid	Focal Tversky Loss	0.7	0.54	0.8

Future Work

While our current work focuses on **building localization using deep learning models,** there are several directions to extend this system toward a **complete disaster assessment framework.**

1. **Advanced Localization Models:** Instead of relying only on U-Net, we plan to explore **SegFormer architectures such as MiT-B2 and MiT-B5,** which use transformer-based feature extraction and can capture **larger spatial context in aerial imagery.**
2. **Hybrid Architectures:** We will also investigate **hybrid CNN–Transformer models** to improve localization accuracy for complex building structures in high-resolution drone imagery.

Once a stronger **building localization model** is developed, the system will move to the **next stage of the pipeline — building damage classification.**

Additional Work

Beyond simple damage detection, the system can also support additional disaster analysis tasks:

Confidence-Aware Predictions: The model can output confidence scores along with damage predictions, allowing low-confidence cases to be flagged for manual inspection by experts.

Class	Probability
No damage	0.08
Damaged	0.92

Additional Work

Area-Level Impact Analysis: By aggregating building-level predictions across regions, the system can provide a high-level overview of disaster impact, helping identify heavily affected areas.

Area/Region	Total Buildings	Damaged Buildings	% Damaged	High Confidence Damaged	Impact Level
Area A	120	45	37.5%	32	High
Area B	80	18	22.5%	10	Medium
Area C	60	5	8.3%	3	Low

Additional Work

Spatial Localization of Damage: Instead of only detecting whether a building is damaged, future models can highlight **which specific parts of the building are affected,** giving responders more detailed structural insights.

These extensions can transform the system from **simple building detection into a comprehensive AI-assisted disaster response tool.**

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Thank You